

# SNAPDRAGON BUILD AND PRESENT CHALLENGE



PROPOSED IDEA : **STUDYSNAP**

An AI-Powered study companion for Snapdragon-Powered PCs

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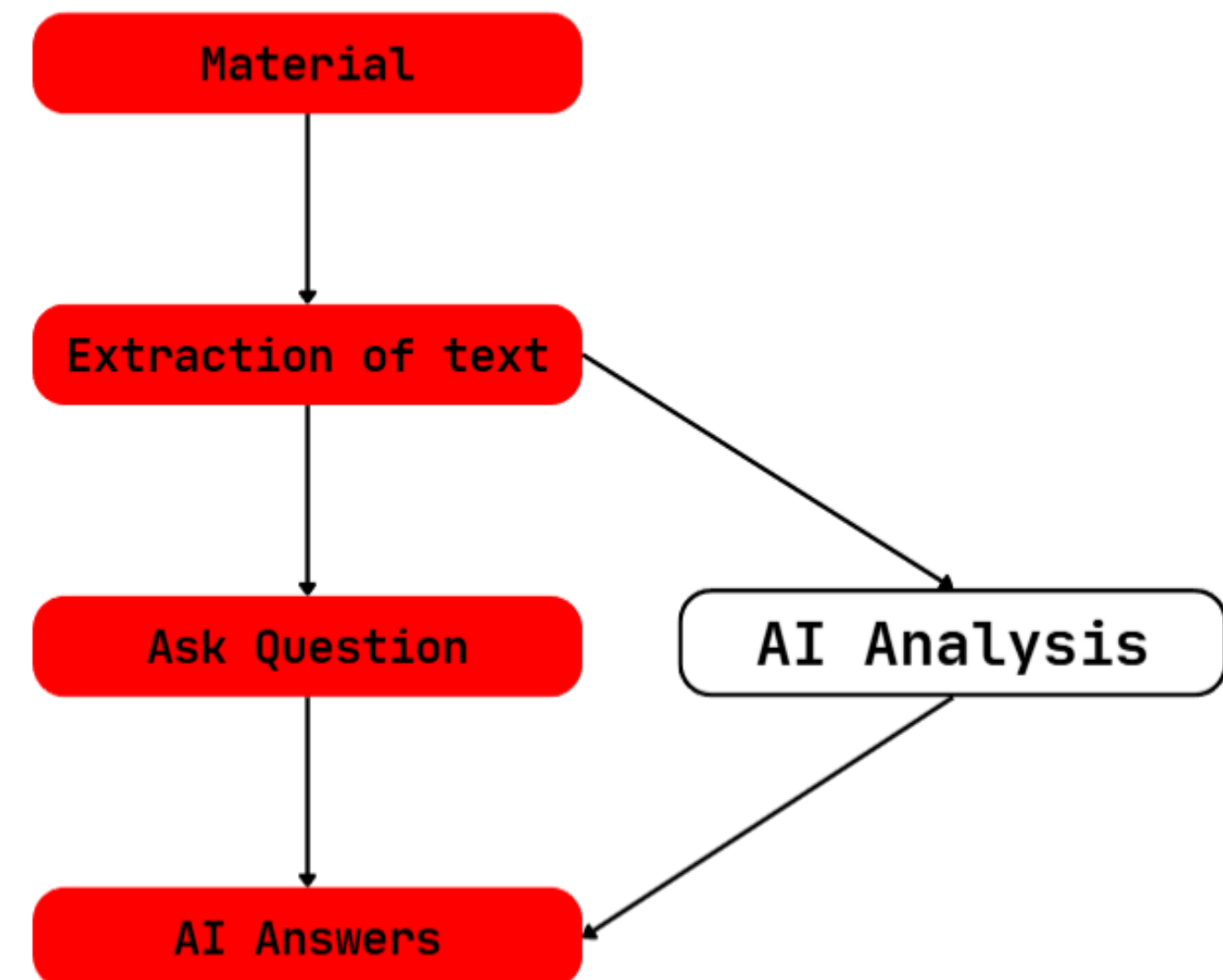
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# THE PROBLEM AND THE SOLUTION

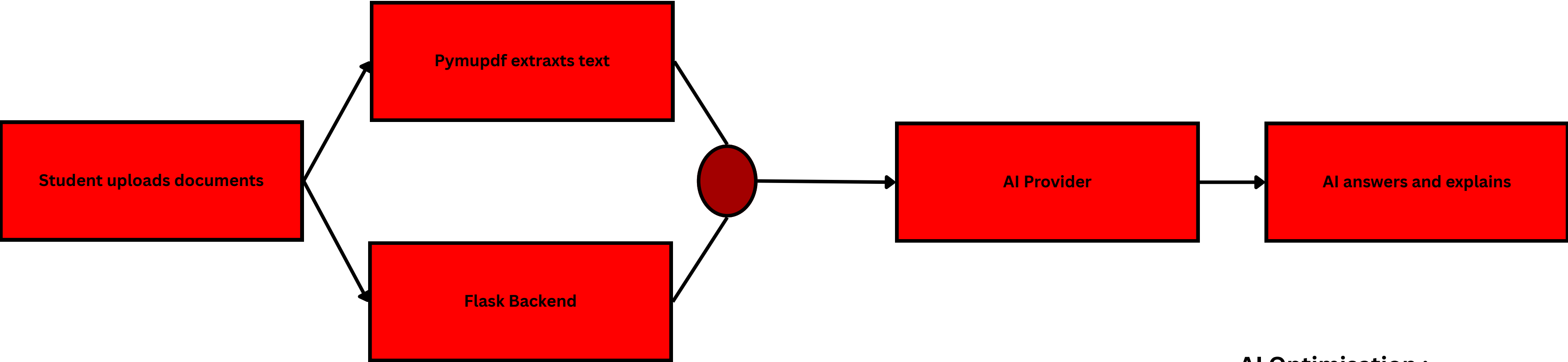
Students normally study from large pdfs and docs but getting useful information is extremely slow.

- Difficult to navigate large pdfs.
- Generic AI tools can't understand students notes.
- Cloud based AI tools have a dependency on the internet.
- Students require explanations rather than just information.

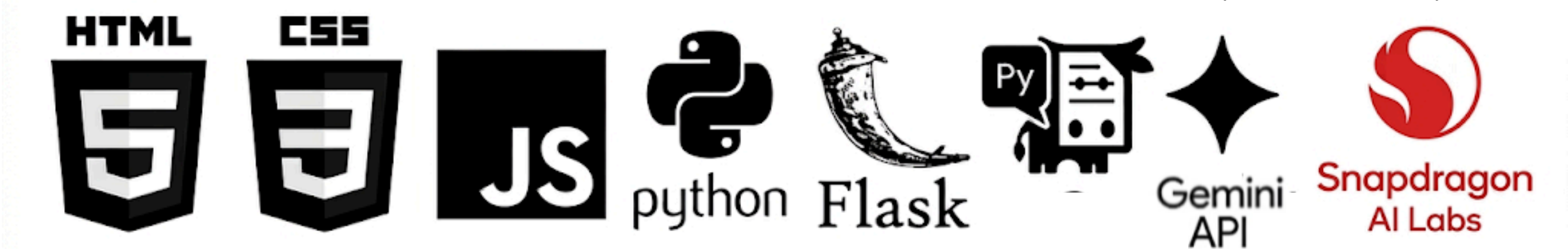
Turns students' material into an interactive one on one study session to ensure proper understanding of concepts.



# HOW STUDYSNAP WORKS



## TECHNOLOGY USED :



AI Optimisation :  
Prototype has been designed with a replaceable AI layer so that it can evolve.

# WHY **SNAPDRAGON**

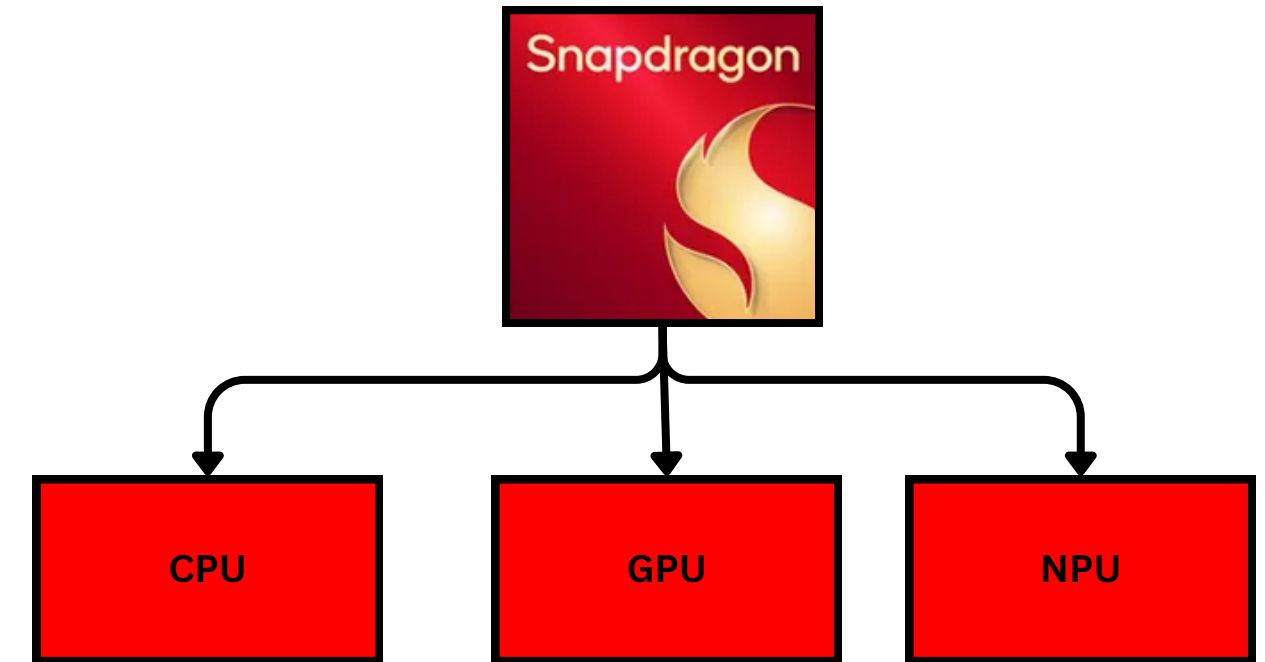
The primary goal is to move the AI assistant from the cloud to the student's PCs.

Snapdragon PCs provide :

CPU : Central Processing unit

GPU : Graphic Processing unit

NPU : Neural Processing unit

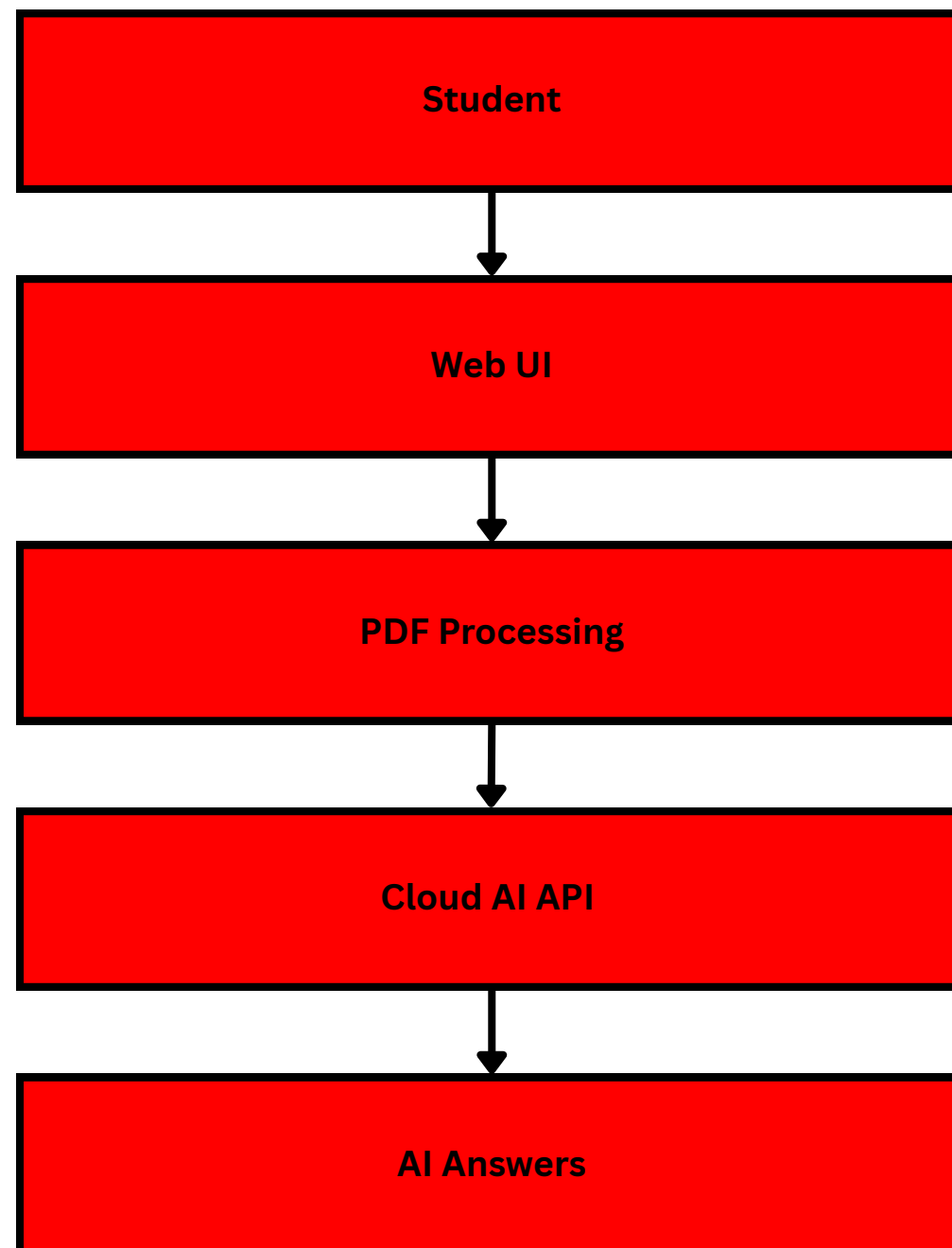


While CPU and GPU are fairly efficient for running AIs, NPU is a unit which has been specifically designed for AI/ML workloads making it the primary hardware target for StudySnap's on-device AI inference.

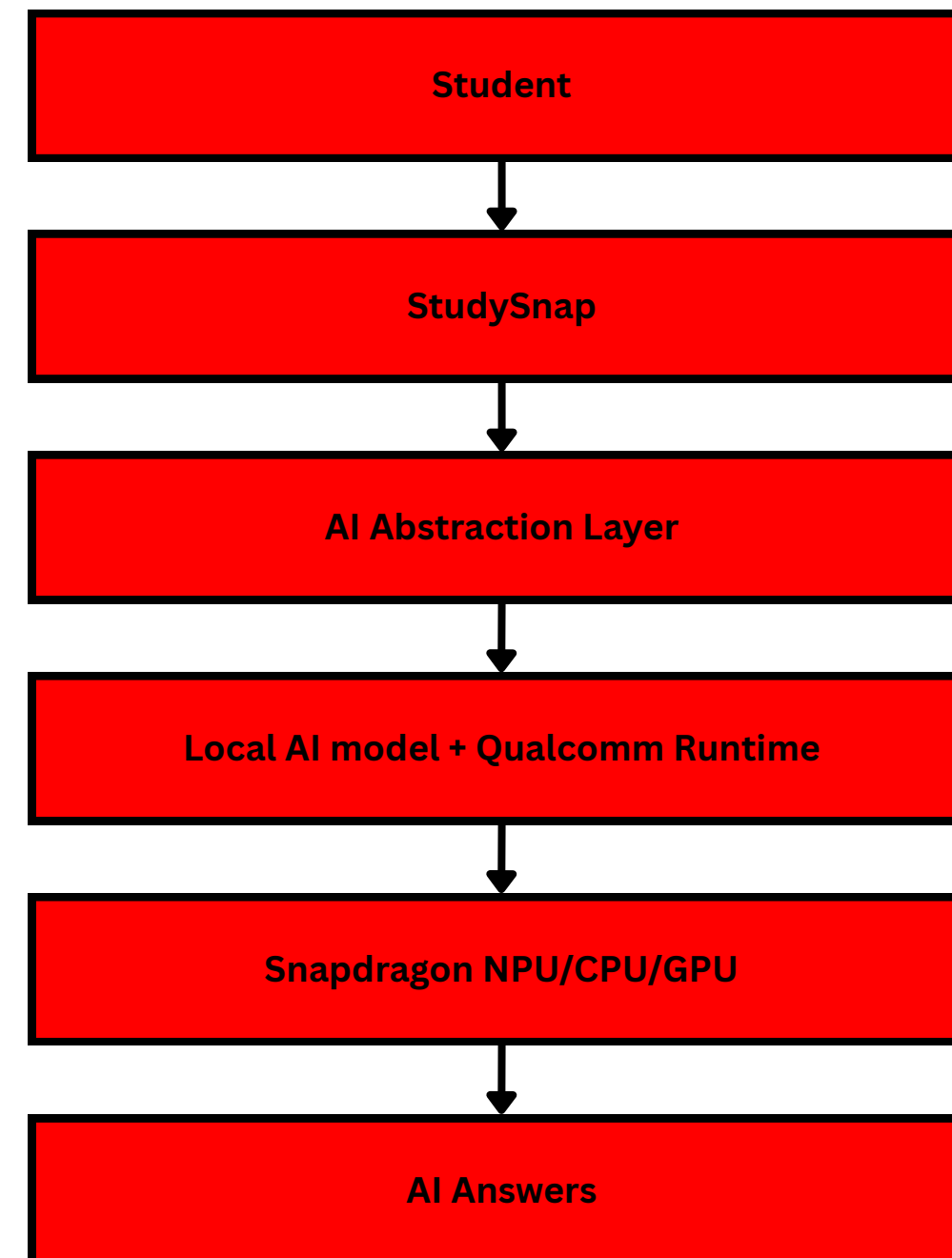
Hexagon NPU : Offers us On-device AI as well as many other features making it well suited for StudySnap.

# CURRENT VS FUTURE ARCHITECTURE

Uses a Cloud API key :



Local AI model and Qualcomm Runtime :



# FUTURE FEATURES, OPTIMISATION AND VISION

## Vision :

“Aims to make AI-Assisted learning more **personal, practical and device-native**. Rather than asking one AI about anything, ask an **AI that understands** what you study directly from your PC.”

## Future Features to be added :

- AI generated quizzes
- Analysing gaps and weak topics
- Adapts to student's learning style
- Voice interaction
- Mind-map generation
- Image/Problem understanding

## Snapdragon Optimisation :

- Define StudySnap requirements.
- Shortlist suitable models
- Profile on qualcomm. supported models.
- Measuring latency, accuracy and NPU footprint.
- Select best suited model.

(Qualcomm AI Hub can support model optimization and profiling workflows.)